Adding and Subtracting Fractions

Name _____ **Date** _____ No calculator Version 1

KEY WORDS numerator · denominator · equivalent fraction · simplest form · common denominator · mixed number · improper fraction

A Equivalent fractions ●○○

ANSWER ON THE SHEET

Remember: Multiply or divide the top and the bottom by the same number: $\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$.

1. Write the missing number in each box.

a) $\frac{1}{2} = \frac{\square}{8}$ **b)** $\frac{2}{3} = \frac{\square}{9}$ **c)** $\frac{3}{4} = \frac{\square}{20}$ **d)** $\frac{2}{5} = \frac{8}{\square}$

e) $\frac{5}{6} = \frac{\square}{12}$ **f)** $\frac{3}{8} = \frac{9}{\square}$ **g)** $\frac{7}{10} = \frac{\square}{100}$ **h)** $\frac{4}{9} = \frac{16}{\square}$

B Simplest form ●○○

ANSWER ON THE SHEET

Watch out: Divide the top and the bottom by the same number, and keep going until no number divides into both.

2. Write each fraction in its simplest form.

a) $\frac{4}{8}$ = _____ **b)** $\frac{6}{9}$ = _____ **c)** $\frac{10}{15}$ = _____ **d)** $\frac{12}{16}$ = _____

e) $\frac{8}{20}$ = _____ **f)** $\frac{15}{25}$ = _____ **g)** $\frac{18}{24}$ = _____ **h)** $\frac{14}{21}$ = _____

i) $\frac{30}{40}$ = _____ **j)** $\frac{24}{36}$ = _____ **k)** $\frac{45}{60}$ = _____ **l)** $\frac{28}{42}$ = _____

C The same denominator ●○○

ANSWER ON THE SHEET

Watch out: Add the numerators only. The denominator stays the same: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$, not $\frac{3}{10}$.

3. Work out. Give each answer in its simplest form.

a) $\frac{1}{5} + \frac{2}{5}$ = _____ **b)** $\frac{3}{7} + \frac{2}{7}$ = _____ **c)** $\frac{5}{9} + \frac{1}{9}$ = _____ **d)** $\frac{7}{8} - \frac{3}{8}$ = _____

e) $\frac{5}{6} - \frac{1}{6}$ = _____ **f)** $\frac{3}{10} + \frac{1}{10}$ = _____ **g)** $\frac{7}{12} - \frac{1}{12}$ = _____ **h)** $\frac{4}{5} + \frac{3}{5}$ = _____

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D Different denominators ●●○

WORK IN YOUR BOOK

Remember: Change both fractions so that they have the same denominator, then add or subtract the numerators.

4. Work out. Give each answer in its simplest form.

a) $\frac{1}{2}+\frac{1}{4}$ **b)** $\frac{1}{3}+\frac{1}{6}$ **c)** $\frac{3}{4}-\frac{1}{8}$ **d)** $\frac{1}{2}+\frac{1}{3}$

e) $\frac{2}{3}+\frac{1}{4}$ **f)** $\frac{4}{5}-\frac{1}{2}$ **g)** $\frac{3}{4}+\frac{2}{5}$ **h)** $\frac{5}{6}-\frac{3}{4}$

i) $\frac{3}{8}+\frac{1}{6}$ **j)** $\frac{7}{10}-\frac{1}{4}$ **k)** $\frac{2}{3}+\frac{3}{5}$ **l)** $\frac{5}{6}-\frac{2}{9}$

E Mixed numbers ●●●

WORK IN YOUR BOOK

Remember: One way is to write each mixed number as an improper fraction first: $1\frac{1}{2}=\frac{3}{2}$.

5. Write each improper fraction as a mixed number, and each mixed number as an improper fraction.

a) $\frac{7}{4}$ **b)** $\frac{11}{3}$ **c)** $2\frac{1}{2}$ **d)** $3\frac{2}{5}$

6. Work out. Give each answer as a mixed number in its simplest form.

a) $1\frac{1}{2}+1\frac{1}{4}$ **b)** $2\frac{1}{3}+1\frac{1}{6}$ **c)** $3\frac{3}{4}-1\frac{1}{2}$ **d)** $1\frac{2}{3}+2\frac{1}{2}$

e) $2\frac{1}{4}-1\frac{1}{2}$ **f)** $1\frac{3}{5}+1\frac{1}{2}$ **g)** $3\frac{1}{3}-1\frac{3}{4}$ **h)** $2\frac{5}{6}+1\frac{2}{3}$

F Problems ●●●

WORK IN YOUR BOOK

7. Ana eats $\frac{1}{4}$ of a pizza and Ben eats $\frac{3}{8}$ of the same pizza. What fraction of the pizza have they eaten altogether?

8. A walk has two parts. The first part is $1\frac{3}{4}$ km long and the second part is $2\frac{1}{2}$ km long. How long is the whole walk, in kilometres?

9. In a class, $\frac{2}{5}$ of the students walk to school and $\frac{1}{4}$ come by bus. The rest come by car. What fraction of the class come by car?

★ Extension

FOR EARLY FINISHERS

10. Work out $\frac{1}{2}+\frac{1}{3}+\frac{1}{6}$.

11. Find the missing fraction: $\frac{1}{2}+\frac{1}{3}+\square=\frac{11}{12}$

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Adding and Subtracting Fractions

Name _____ **Date** _____ No calculator Version 2

KEY WORDS numerator · denominator · equivalent fraction · simplest form · common denominator · mixed number · improper fraction

A Equivalent fractions ●○○

ANSWER ON THE SHEET

Remember: Multiply or divide the top and the bottom by the same number: $\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$.

1. Write the missing number in each box.

a) $\frac{1}{3} = \frac{\square}{12}$ **b)** $\frac{3}{4} = \frac{\square}{12}$ **c)** $\frac{2}{5} = \frac{\square}{25}$ **d)** $\frac{3}{5} = \frac{12}{\square}$

e) $\frac{5}{8} = \frac{\square}{16}$ **f)** $\frac{4}{7} = \frac{12}{\square}$ **g)** $\frac{9}{10} = \frac{\square}{100}$ **h)** $\frac{5}{9} = \frac{20}{\square}$

B Simplest form ●○○

ANSWER ON THE SHEET

Watch out: Divide the top and the bottom by the same number, and keep going until no number divides into both.

2. Write each fraction in its simplest form.

a) $\frac{3}{6}$ = _____ **b)** $\frac{8}{12}$ = _____ **c)** $\frac{10}{25}$ = _____ **d)** $\frac{12}{18}$ = _____

e) $\frac{6}{20}$ = _____ **f)** $\frac{20}{25}$ = _____ **g)** $\frac{16}{24}$ = _____ **h)** $\frac{21}{28}$ = _____

i) $\frac{30}{50}$ = _____ **j)** $\frac{27}{36}$ = _____ **k)** $\frac{45}{75}$ = _____ **l)** $\frac{32}{48}$ = _____

C The same denominator ●○○

ANSWER ON THE SHEET

Watch out: Add the numerators only. The denominator stays the same: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$, not $\frac{3}{10}$.

3. Work out. Give each answer in its simplest form.

a) $\frac{2}{7} + \frac{3}{7}$ = _____ **b)** $\frac{1}{9} + \frac{4}{9}$ = _____ **c)** $\frac{3}{8} + \frac{3}{8}$ = _____ **d)** $\frac{7}{10} - \frac{3}{10}$ = _____

e) $\frac{7}{9} - \frac{4}{9}$ = _____ **f)** $\frac{5}{12} + \frac{1}{12}$ = _____ **g)** $\frac{9}{10} - \frac{3}{10}$ = _____ **h)** $\frac{5}{6} + \frac{5}{6}$ = _____

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D Different denominators ●●○

WORK IN YOUR BOOK

Remember: Change both fractions so that they have the same denominator, then add or subtract the numerators.

4. Work out. Give each answer in its simplest form.

a) $\frac{1}{2}+\frac{1}{8}$ **b)** $\frac{1}{4}+\frac{1}{12}$ **c)** $\frac{5}{6}-\frac{1}{3}$ **d)** $\frac{1}{2}+\frac{1}{5}$

e) $\frac{3}{4}+\frac{1}{3}$ **f)** $\frac{3}{5}-\frac{1}{2}$ **g)** $\frac{2}{3}+\frac{2}{5}$ **h)** $\frac{7}{8}-\frac{2}{3}$

i) $\frac{5}{8}+\frac{1}{6}$ **j)** $\frac{9}{10}-\frac{3}{4}$ **k)** $\frac{3}{4}+\frac{4}{5}$ **l)** $\frac{7}{9}-\frac{1}{6}$

E Mixed numbers ●●●

WORK IN YOUR BOOK

Remember: One way is to write each mixed number as an improper fraction first: $1\frac{1}{2}=\frac{3}{2}$.

5. Write each improper fraction as a mixed number, and each mixed number as an improper fraction.

a) $\frac{9}{4}$ **b)** $\frac{13}{3}$ **c)** $3\frac{1}{2}$ **d)** $2\frac{3}{5}$

6. Work out. Give each answer as a mixed number in its simplest form.

a) $1\frac{1}{4}+2\frac{1}{2}$ **b)** $1\frac{1}{3}+2\frac{1}{6}$ **c)** $3\frac{5}{6}-1\frac{1}{3}$ **d)** $2\frac{1}{2}+1\frac{2}{3}$

e) $3\frac{1}{4}-1\frac{1}{2}$ **f)** $1\frac{4}{5}+1\frac{1}{2}$ **g)** $4\frac{1}{3}-2\frac{3}{4}$ **h)** $1\frac{5}{6}+2\frac{2}{3}$

F Problems ●●●

WORK IN YOUR BOOK

7. Ana eats $\frac{1}{3}$ of a pizza and Ben eats $\frac{2}{9}$ of the same pizza. What fraction of the pizza have they eaten altogether?

8. A walk has two parts. The first part is $2\frac{1}{4}$ km long and the second part is $1\frac{1}{2}$ km long. How long is the whole walk, in kilometres?

9. In a class, $\frac{1}{5}$ of the students walk to school and $\frac{1}{2}$ come by bus. The rest come by car. What fraction of the class come by car?

★ Extension

FOR EARLY FINISHERS

10. Work out $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}$.

11. Find the missing fraction: $\frac{1}{2}+\frac{1}{4}+\square=\frac{7}{8}$

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